

Ziyi Sun

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College of Engineering, Ocean University of China, Qingdao 266100, China

EDUCATION

• Ocean University of China (985 / Double First-Class)

Sep. 2023 – Jun. 2027 (Expected)

B.Eng. in Mechanical Design, Manufacturing and Automation

Qingdao, China

◦ GPA: **3.7/4.0** Rank: **1/56** Average: 89.2

◦ Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

• Roboparty Lab

Jul. 2026 – Oct. 2026

Frontier Research and Open Source Intern (Mentor: Jagger)

Beijing, China

◦ Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

PROJECTS

• Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection

Dec. 2025 – Dec. 2026

Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing

OUC SRDP

◦ Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.

◦ Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.

• Cross-Domain Wire-Driven Biomimetic Eel Robot

Dec. 2024 – Nov. 2025

Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection

Provincial Innovation Program

◦ Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.

◦ Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.

• Streaming Evaluation Framework for Deep-Sea Mining

2025 – 2026

Role: Lead Contributor | Tools: CAD modeling, multi-AUV simulation & benchmarking

Research Project

◦ Built a unified simulation and benchmarking pipeline comparing centralized and distributed multi-AUV nodule collection strategies.

◦ Integrated CAD-based robot models and reproducible evaluation protocols for scalable underwater robot learning.

• High-Stability Duck-Type Wave Energy Converter with Pendulum PTO

Dec. 2024 – Nov. 2025

Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks

National Innovation Program

◦ Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

PUBLICATIONS

C=CONFERENCE, S=IN SUBMISSION

[C.1] Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.
A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.
IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist.** [PDF]

[C.2] Ziyi Sun et al.
Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.
IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.

[S.1] Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.
CHEROE: Every Humanoid Skill as a Traject.
Under review at AAAI 2027 (CCF-A).

[S.2] Ziyi Sun, Changhong He.
A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.
Under review at Journal of Marine Science and Engineering (JMSE).

PATENTS & SOFTWARE

- **Invention Patent:** A High-Stability PTO System for Duck-Type Wave Energy Converters (First Student Inventor)
- **Software Copyright:** Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0 (No. 2024SR0140235)

SKILLS

- **Robot Platform:** Unitree G1
- **Programming & ML:** Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:** SolidWorks, AutoCAD
- **Research Focus:** Embodied AI, robot learning, sim-to-real transfer, multimodal perception, underwater robotics

HONORS AND AWARDS

- **Shandong Provincial Government Scholarship** 2025
- **First-Class Comprehensive Scholarship, Ocean University of China** 2024 & 2025
- **Outstanding Student / Outstanding Student Cadre, Ocean University of China** 2024 & 2025
- **National First Prize, National 3D Digital Innovation Design Competition** 2025 & 2026
- **National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)** 2025
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** 2025
- **National Third Prize, 18th National College Student Energy Conservation Competition** 2024
- **National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest** 2024
- **National Second Prize, 14th Marine Vehicle Design and Production Competition** 2025
- **National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition** 2026
- **Science and Technology Innovation Pioneer Team, College of Engineering, OUC** 2024

LEADERSHIP EXPERIENCE

- **Class Representative / Class Monitor** 2023 – Present
Ocean University of China
 - Coordinated class affairs and served as the primary liaison between students and faculty.
- **Team Leader, College of Engineering Debate Team** 2023 – Present
Ocean University of China
 - Led training and competition preparation for the college debate team.
- **Director of Event Organization** 2023 – Present
Student Association for the Study of Socialism with Chinese Characteristics, OUC
 - Organized association events and coordinated student outreach activities.

VOLUNTEER EXPERIENCE

- **Volunteer** 2023
Qingdao Marathon
 - Supported on-site volunteer operations for a major city marathon event.
- **Member, National College Student Volunteer Outreach Team** 2024
Promoting the Spirit of the Zunyi Conference
 - Participated in national college student volunteer outreach and public education activities.
- **Outstanding Individual in Summer Social Practice** 2024
Ocean University of China
 - Recognized for leadership and contribution during summer social practice.

ADDITIONAL INFORMATION

Languages: Chinese (Native); English (CET-4: 591, CET-6: 482)

Interests: Embodied intelligence, robot learning, underwater robotics, human–robot interaction